

Course: NE 533 Nuclear Fuel Performance (Beeler)-- Exam 1STUDENT'S PRINTED NAME: Ingrid Caraballo López**INSTRUCTIONS FOR ADMINISTRATION:**

- To be taken **between Tuesday, February 3 and Wednesday, February 11 , 2026.**
- Total Pages (including this cover): **Three (3)**
- Due to Engineering Online office: **end of day (11:59 pm Eastern) Wednesday, February 11, 2026.**
- Time limit: **75 minutes**
- **Specific Instructions:** Closed book, Closed notes, Graphing calculator allowed, No computer/internet access allowed **Equation sheet will be provided with exam. DO NOT WRITE ON THE BACK OF ANY PAGES. Additional paper needed to complete exam.**
- **Additional prohibited:** PDAs, cell phones, pagers, iPods, digital media players, text messaging devices, or other electronic devices not expressly permitted above.

THE FOLLOWING HONOR STATEMENT AND TIME RECORDING MUST BE COMPLETED:**Student - NC State University Honor Pledge:**

I, Ingrid Caraballo López, have neither given nor received unauthorized aid on this exam or assignment.
(STUDENT'S PRINTED NAME)

Student Signature

Date

Proctor:

Time Student Began Exam: 11:00 Time Student Completed Exam: 12:50 Total Time: 1hr 50min

I, Dana Wynn, administered this exam according to the instructions above.
(PROCTOR'S PRINTED NAME)

Proctor Signature

Date

RETURN INSTRUCTIONS: The completed exam (questions, answers, any work sheets/extra pages, and this form) **MUST ALL** be returned **BY THE PROCTOR** via email or fax to Engineering Online **immediately** upon completion.

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Exam 1: NE533: Nuclear Fuel Performance

Show all work. Label question number in your response. Pay attention to units. Point values correspond to expected depth of response.

1. (12 pts) UN is a fuel being considered for use in light water reactors. Consider UN to have an enrichment of 19% and a density of 14.3 g/cm^3 . Assume the fission cross section is 587 barns. Nitrogen mass number=14.

What is the heat generation rate, given a neutron flux of $4 \times 10^{12} \text{ n/cm}^2\text{-s}$? What enrichment of UO_2 would be required to obtain the same heat generation rate? UO_2 density is 10.97 g/cc .

2. (18 pts) Calculate the temperature at the radial midpoint of the cladding and at $r=0.1 \text{ cm}$. Assume gap is all He.

Cladding k : 0.18 W/cm-K ; Fuel k : 0.04 W/cm-K ; Coolant h = $1.5 \text{ W/cm}^2\text{-K}$; Q = 350 W/cm^3 ; $R_F=0.45 \text{ cm}$; $t_g=80 \text{ }\mu\text{m}$; $t_{\text{clad}}=0.06 \text{ cm}$; $T_{\text{cool}} = 540 \text{ K}$

3. (8 pts) Given a rod of 3.6 m in length, $\text{LHR}^0 = 400 \text{ W/cm}$, and $\gamma=1.25$:

What is the LHR at $z=2.1 \text{ m}$? What is the coolant outlet temperature? Assume water: $C_p = 4200 \text{ J/kg-K}$, $\dot{m} = 0.1 \text{ kg/s-rod}$, inlet temperature = 580 K .

4. (6 pts) By how much does the thermal conductivity of UO_2 degrade at a burnup of 5% FIMA? Assume an average fuel temperature of 1200 K .

5. (6 pts) What assumptions to the heat conduction equation were required to derive the analytical form of the temperature profile?
6. (5 pts) What compound is utilized in the enrichment process? Describe the centrifuge-based enrichment of U, including why it works.
7. (8 pts) Discuss the departure from nucleate boiling ratio and the critical heat flux.
8. (4 pts) What is smear density? Why is a reduced smear density necessary?
9. (4 pts) What are two sources of thermal conductivity degradation in UO_2 ?
10. (6 pts) What is the role of cladding? Why is Zr suitable as a cladding material?
11. (4 pts) What does the "fuel system" consist of?
12. (4 pts) What are the three aspects that I define as constituting fuel performance?
13. (6 pts) What are the three ways that space is discretized for numerical solutions? Name a strength or weakness of one of these types.
14. (5 pts) Name two positive and two negative aspects of UO_2 fuel.
15. (4 pts) What is the conceptual difference between implicit and explicit time integration?

1. U-N

$$\epsilon_{UN} = 19\%$$

$$\epsilon_{U235} = 0.81$$

$$\rho_{UN} = 14.3 \text{ g/cm}^3$$

$$\sigma_f = 587 \text{ barns}$$

$$\phi = 4 \times 10^{12} \text{ n/cm}^2\text{-s}$$

$$1 \text{ barn} = 10^{-24} \text{ cm}^2$$

$$\epsilon_{U235} = \text{UN} = 9.54 + 0.5N$$

$$\rho_{UO_2} = 10.97 \text{ g/cm}^3$$

$$N_A = 6.02 \times 10^{23} \text{ moles/cm}^3$$

$$E_f = 200 \times 10^6 \text{ eV} \times 1.60 \times 10^{-19} \left[\frac{\text{J}}{\text{eV}} \right]$$

* Assuming $X_{U235} = 0.97$ $X_N = 0.03$
I don't see the molar values for it

① Heat generation rate

a) molecular weight

$$m_{(UN)} = X_{U235} (\epsilon_{U235} \times A_{U235} + \epsilon_{U238} \times A_{U238}) + X_N \times A_N$$

$$m_{(UN)} = 0.97 (0.81 (235) + 0.19 (238)) + (0.03)(14)$$

$$m_{(UN)} = 214.789 \text{ g/mol}$$

b) Number density $N_f(UN)$

$$N_f(UN) [\text{mole/cm}^3] = \frac{\rho_{UN} \times N_A}{m_{W(UN)}} =$$

$$\frac{14.3 [\text{g/cm}^3] \times 6.02 \times 10^{23} \frac{\text{mole}}{\text{cm}^3}}{214.789 \text{ g/mol}} = 4.007 \times 10^{22}$$

c) Number density $N_f(U235)$

$$N_f(U235) [\text{atoms/cm}^3] = N_f(UN) \times X_{U235} \times \epsilon_{U235}$$

$$N_f(U235) = (4.007 \times 10^{22}) (0.97) (0.81) = 3.149 \times 10^{22}$$

d) Heat generation $Q = E_f \times N_f \times \sigma_f \times \phi$

$$Q = (200 \times 10^6 \text{ eV} \times 1.60 \times 10^{-19} \left[\frac{\text{J}}{\text{eV}} \right]) \times (3.149 \times 10^{22}) \times (587 \times 10^{-24} \text{ cm}^2) (4 \times 10^{12} \text{ n/cm}^2\text{-s})$$

$$Q = 2366.04 \text{ - super high!}$$

enrichment of UO_2 required * substituting N_f expressed for ϵ_{U235} in heat gen eq.

$$\epsilon_{U235} = \frac{Q (X_{U235} + A_{U235} + X_N \times A_N)}{E_f \times \sigma_f \times \phi \times \rho_{U235} \times X_{U235} - Q \times X_{U235} + (A_{U235} - A_{U238})}$$

- didn't see how you got here..

- can't tell if

this is correct

$$\epsilon_{U235} = \frac{2366.04 (0.97 + 235 + 0.03 \times 14)}{(200 \times 10^6 \text{ eV} \times 1.60 \times 10^{-19} \left[\frac{\text{J}}{\text{eV}} \right]) (587 \times 10^{-24} \text{ cm}^2) (4 \times 10^{12} \text{ n/cm}^2\text{-s}) (10.97 \text{ g/cm}^3) (0.97) - 2366.04 (0.97) + (235 - 238)}$$

$$\epsilon_{U235} = 0.92\%$$

→ should be higher than UN enrichment... 2/7

2. Temperature at radial midpoint

at $r = 0.1 \text{ cm}$

$$K_{\text{clad}} = 0.18 \text{ W/cm-K} \quad t_{\text{clad}} = 0.06 \text{ cm}$$

$$K_{\text{fuel}} = 0.04 \text{ W/cm-K}$$

$$K_{\text{cool}} = ? \quad T_{\text{cool}} = 540 \text{ K}$$

$$h_{\text{cool}} = 1.5 \text{ W/cm}^2\text{-K}$$

$$Q = 350 \text{ W/cm}^3$$

$$R_f = 0.45 \text{ cm}$$

$$h_{\text{gap}} = ? \quad t_{\text{gap}} = 80 \mu\text{m} \quad K_{\text{gap}}(1\text{K}) = 16 \times 10^{-6} \times T^{0.71}$$

$$\text{LHR} = \pi R_f^2 Q_{\text{av}}$$

$$\text{LHR} = \pi (0.45 \text{ cm}) (350 \text{ W/cm}^3)$$

$$\text{LHR} = 222.66$$

$$\textcircled{A} T_{\text{oc}} - T_{\text{cool}} = \frac{\text{LHR}}{2\pi R_f h_{\text{cool}}}$$

outer clad

$$T_{\text{oc}} = \frac{222.66}{2\pi (0.45 \text{ cm}) (1.5 \text{ W/cm}^2\text{-K})} + 540 \text{ K} = 592.5 \text{ K}$$

$$\textcircled{B} T_{\text{ic}} - T_{\text{oc}} = \frac{\text{LHR}}{2\pi R_f h_{\text{clad}}}$$

inner clad

$$h_{\text{clad}} = ? = \frac{K_{\text{clad}}}{t_{\text{clad}}} = \frac{0.18}{0.06} = 3 \text{ W/cm}^2\text{-K}$$

$$T_{\text{ic}} = \frac{222.66}{2\pi (0.45 \text{ cm}) (3 \text{ W/cm}^2\text{-K})} + 592.5 \text{ K} = 618.75 \text{ K}$$

$$\textcircled{C} T_s - T_{\text{ic}} = \frac{\text{LHR}}{2\pi R_f h_{\text{gap}}}$$

surface

$$t_{\text{gap}} = 0.008 \text{ cm}$$

$$h_{\text{gap}} = ?$$

assume He

$$K_{\text{gap}}(\text{He}) = 16 \times 10^{-6} \times (T_{\text{ic}})^{0.79}$$

$$K_{\text{gap}}(1\text{K}) = 16 \times 10^{-6} (618.75 \text{ K})^{0.79} = 0.002567 \text{ W/cm-K}$$

$$h_{\text{gap}} = \frac{K_{\text{gap}}}{t_{\text{gap}}} = \frac{0.002567}{0.008 \text{ cm}} = 0.320875 \text{ W/cm}^2\text{-K}$$

$$T_s = \frac{222.66}{2\pi (0.45 \text{ cm}) (0.320875 \text{ W/cm}^2\text{-K})} + 618.75 \text{ K} = 864.172 \text{ K}$$

$$\textcircled{1} T_o - T_s = \frac{LHR}{4\pi k_f} \quad \text{centerline}$$

$$T_o = \frac{222.66}{4\pi (0.04 \text{ W/cm-K})} + 864.172 \text{ K} = 11848.55 \text{ K}$$

- asked for cladding midpoint.

- calculator error?

too high!

- asked for T_{cool}

$$\textcircled{2} r = 0.1 \text{ cm}$$

$$L = 3.6 \text{ m}$$

$$z_o = \frac{3.6 \text{ m}}{2} = 1.8 \text{ m}$$

$$LHR^o = 400 \text{ W/cm} \quad z = 2.1 \text{ m}$$

$$LHR^o = 400 \frac{\text{W}}{\text{cm}} \times \frac{1 \text{ cm}}{15^2 \text{ m}} = 40000 \frac{\text{W}}{\text{m}}$$

$$\gamma = 1.25$$

$$\dot{m} = 0.1 \text{ kg/s-rod}$$

$$C_p = 4200 \text{ J/kg-K}$$

$$T_{cool}^{in} = 580 \text{ K}$$

$T_{cool}(z)$ and LHR ?

$$\textcircled{A} LHR\left(\frac{z}{z_o}\right) = LHR^o \cos\left[\frac{\pi}{2\gamma}\left(\frac{z}{z_o} - 1\right)\right]$$

$$LHR = 40000 \frac{\text{W}}{\text{m}} \times \cos\left[\frac{\pi}{2(1.25)}\left(\frac{2.1}{1.8} - 1\right)\right]$$

$$LHR = 3912.59 \text{ W/m} \quad \checkmark$$

$$\textcircled{B} T_{cool}(z) = \frac{2\gamma}{\pi} \frac{z_o \times LHR^o}{\dot{m} C_p} \left\{ \sin \frac{\pi}{2\gamma} + \sin \left[\frac{\pi}{2\gamma} \left(\frac{z}{z_o} - 1 \right) \right] \right\} + T_{cool}^{in}$$

$$T_{cool}(z) = \frac{2(1.25)}{\pi} \frac{(1.8 \text{ m})(40000 \frac{\text{W}}{\text{m}})}{0.1 [\text{kg/s-rod}] 4200 \text{ J/kg-K}} \left\{ \sin \frac{\pi}{2(1.25)} + \sin \left[\frac{\pi}{2(1.25)} \left(\frac{2.1 \text{ m}}{1.8 \text{ m}} - 1 \right) \right] \right\}$$

$$+ 580 \text{ K} = 738.105 \text{ K}$$

- coolant @ outlet!

$$z = 3.6$$

$$T_{cool}(z) = 738.105 \text{ K}$$

$$\textcircled{4} K_{ox} = \frac{1}{A + BT}$$

$$A = 3.8 + 200 \times FIMA$$

$$A = 3.8 + 200(0.05) = 13.8$$

$$B = 0.0217 \text{ (cm/W)}$$

$$K_{ox} = \frac{1}{13.8 + 0.0217(1200 \text{ K})}$$

- how much originally?

- need reference point

$$K_{ox} = 0.0251$$

degrade

5. Assumptions to the heat conduction eq.

1. Steady state ✓
2. Neglect axial heat conduction (cylindrical) asymmetry ✓
3. Uniform volumetric heat generation ✓
4. Constant thermal conductivity ✓

6. Compounds in enrichment process

U_3O_8 must be converted to UF_6 for enrichment which is then converted to UO_2 powder for pellet manufacturing. (sintering) ✓

3/5 Centrifuge works because the particles are violently pushed through a gaseous process while the unwanted product is removed from the fuel. This is determined through the SWU value. → not sufficient

7. CHF and DNBR

5/8 Is the heat at which boiling ceases to be effective form of transferring heat from solid surface to liquid. As the coolant temp. increases the mode of heat transfer changes. While DNBR is above the nucleate boiling, where the surface is no longer hotter than the saturated fluid temperature. - not quite on DNBR

8. Smear density is the ratio of fuel to volume to total internal

volume of the cladding. ✓

3/4 reduced smearing is necessary to account for the necessary gap required in the fuel. Why is a gap needed?

9. Thermal conductivity degradation in UO_2

Corrosion is a source and the porosity in the powdered fuel when compacted.

1/4 - UO_2 is not corroded - initial porosity is initial, not temporal

10. Role of cladding

Cladding separates the coolant from the fuel. ✓

2/6 Fuel pins are made with zircalloy because it has a low neutron cross section ✓ and has resistance to corrosion and swelling. While it also has good thermal conductivity. ✓

11. The fuel system consist of the cladding, pellets, spring and adjunctions that are held and named a fuel pin.

3/4 - gap, coolant

12. The three aspects are

- manufacturing of the fuel, how it is made X
- the system it is used in, temperature and all X
- how it performs under these conditions, heat released ✓

13. By time, burnup and geometry

1/6 stiff equations are one and are numerically unstable, unless the step is taken to be extremely small. - space! F_g, F_v, F_E

14. UO_2 has

Good - melting point ✓

- compatible with zircaloy ✓

Bad - Brittle ✓

- poor thermal conductivity ✓

15. Explicit: calculates the state at which a system
at a later time while implicit finds a solution
involving both of the current and later states

Equations for NE533

$$\begin{aligned}
 Q &= E_f \times N_f \times \sigma_f \times \phi \\
 \rho c_p \frac{\partial T}{\partial t} &= \nabla \cdot (k \nabla T) + Q \\
 LHR &= \pi R^2 Q_{av} \\
 T(r) &= \frac{Q(R^2 - r^2)}{4k} + T_s \\
 T_0 - T_s &= \frac{Q}{4k} R_f^2 = \frac{LHR}{4\pi k} \\
 T_s - T_{IC} &= \frac{Q}{2h_{gap}} R_f = \frac{LHR}{2\pi R_f h_{gap}} \\
 T_{IC} - T_{OC} &= \frac{Q}{2h_{clad}} R_f = \frac{LHR}{2\pi R_f h_{clad}} \\
 T_{OC} - T_{cool} &= \frac{Q}{2h_{cool}} R_f = \frac{LHR}{2\pi R_f h_{cool}} \\
 h_{gap/clad} &= \frac{k_{gap/clad}}{t_{gap/clad}} \\
 k_{gap}(He) &= 16 \times 10^{-6} \times T^{0.79} \\
 k_{gap}(Xe) &= 0.7 \times 10^{-6} \times T^{0.79} \\
 k_{gap}(y) &= (1 - y) \times k_{He} + y \times k_{Xe} \\
 LHR \left(\frac{z}{Z_0} \right) &= LHR^0 \cos \left[\frac{\pi}{2\gamma} \left(\frac{z}{Z_0} - 1 \right) \right] \\
 T_{cool}(z) - T_{cool}^{in} &= \frac{2\gamma}{\pi} \frac{Z_0 \times LHR^0}{\dot{m} C_p} \left\{ \sin \frac{\pi}{2\gamma} + \sin \left[\frac{\pi}{2\gamma} \left(\frac{z}{Z_0} - 1 \right) \right] \right\} \\
 k_{ox} &= \frac{1}{A + BT} \text{ (W/cmK)} \\
 \frac{1}{B} \ln \left(\frac{A + BT_0}{A + BT_s} \right) &= \frac{LHR}{4\pi} \\
 A &= 3.8 + 200 \times FIMA \text{ (cmK/W)} \\
 B &= 0.0217 \text{ (cm/W)}
 \end{aligned}$$